

TruckOrder

Sandwich Stall Ordering Portal

Product Requirements Document (PRD)

Version 1.0 · March 2026

1. Product Overview

TruckOrder is a real-time web portal built with React and TypeScript for a sandwich stall food truck. It enables staff to monitor and manage customer orders across all lifecycle stages from a single, live-updating dashboard — eliminating paper order slips and manual tracking.

Product Name	TruckOrder — Sandwich Stall Portal
Version	1.0
Platform	React 18 + TypeScript (Web)
Target Users	Sandwich stall staff & managers
Document Date	March 2026
Status	Draft — Ready for Development

2. Goals & Success Metrics

2.1 Business Goals

- Reduce order errors by providing a single source of truth for all active orders
- Speed up fulfilment by giving kitchen staff real-time order visibility
- Enable managers to track revenue and order volume without manual tally
- Provide a scalable foundation that can be extended to multiple stalls

2.2 Success Metrics

Metric	Target	Measurement
Order accuracy rate	≥ 98%	Monthly review
Dashboard load time	< 2 seconds	Lighthouse score
Real-time update latency	< 500 ms	WebSocket ping
Staff onboarding time	< 15 minutes	User testing

3. Users & Roles

Role	Responsibilities	Portal Access
Kitchen Staff	Prepare orders, update status	View orders, change status (New → In Progress → Completed)
Cashier / Front Staff	Take orders from customers	Create new orders, view all orders
Manager / Owner	Monitor revenue and performance	Full dashboard access, order history, totals

4. Menu — Orderable Items

The portal supports exactly 5 menu items at launch. Each item has a fixed price, name, description, and unique ID.

#	Item Name	Description	Price	Category
1	Classic Club	Turkey, ham, bacon, lettuce, tomato, mayo on toasted white	\$9.99	Signature
2	BBQ Brisket Melt	Slow-cooked brisket, cheddar, caramelised onion, BBQ sauce on brioche	\$11.99	Signature
3	Veggie Stack	Roasted peppers, hummus, avocado, rocket, feta on multigrain	\$9.49	Vegetarian

#	Item Name	Description	Price	Category
4	Spicy Chicken	Crispy chicken, sriracha slaw, pickled jalapeños, chipotle mayo on sesame roll	\$10.49	Signature
5	BLT Deluxe	Thick-cut bacon, heirloom tomato, gem lettuce, herb aioli on sourdough	\$8.99	Classic

5. Feature Requirements

5.1 Real-Time Order Dashboard

The main view is a live dashboard that auto-updates without a manual page refresh. It uses WebSockets (or polling fallback every 5 seconds) to push order changes to all connected clients instantly.

- Four status lanes displayed as kanban-style columns or tabbed views
- Each order card shows: Order ID, customer name, items ordered, total price, timestamp, and current status
- Orders animate into the correct column when status changes
- Sound notification (optional toggle) when a new order arrives

5.2 Order Status Lifecycle

Status	Badge Color	Description
New	Blue	Order just placed; awaiting kitchen acknowledgement
In Progress	Amber	Kitchen has started preparing the order
Completed	Green	Order is ready for collection or delivered
Cancelled	Red	Order cancelled before preparation began

Valid state transitions:

- New → In Progress (kitchen accepts the order)
- In Progress → Completed (order ready)
- New → Cancelled (before kitchen starts)
- No backward transitions are permitted once an order is In Progress

5.3 Summary Metrics Bar

A persistent header bar displays live aggregate statistics updated in real time:

Metric Card	Description
Total Orders Today	Count of all orders placed since midnight
New Orders	Count of orders currently in 'New' status
In Progress	Count of orders currently being prepared
Completed Today	Count of fulfilled orders since midnight
Total Revenue Today	Sum of all completed order totals (£ or \$)

5.4 New Order Creation

Front-of-house staff can place a new order directly from the portal:

- Select one or more items from the 5-item menu
- Specify quantity per item (1–10)
- Enter an optional customer name or order note
- Review order total before submitting
- On submission, the order appears instantly in the 'New' column for all clients

5.5 Order Detail View

Clicking any order card opens a side panel or modal with full order details:

- List of all items with quantities and individual prices
- Subtotal and grand total
- Timestamps for each status change (audit trail)
- Action buttons to advance or cancel the order

6. Non-Functional Requirements

Category	Requirement	Acceptance Criteria
Performance	Dashboard load time	< 2 seconds on 4G mobile
Real-Time	Order update latency	< 500 ms end-to-end via WebSocket
Responsiveness	Mobile & tablet support	Fully usable on screens ≥ 360 px wide
Reliability	Uptime during service hours	≥ 99.5% during 7 am – 9 pm
Scalability	Concurrent users	Handles 20 simultaneous portal users

Category	Requirement	Acceptance Criteria
Browser Support	Cross-browser	Chrome, Firefox, Safari, Edge (latest 2 versions)
Accessibility	WCAG 2.1 AA	Colour contrast $\geq 4.5:1$, keyboard navigable

7. Technical Architecture

7.1 Frontend Stack

Layer	Technology	Purpose
UI Framework	React 18	Component-based UI rendering
Language	TypeScript 5+	Type safety across all components
State Management	Zustand or React Context	Global order state and real-time updates
Styling	Tailwind CSS	Utility-first responsive design
Real-Time	WebSocket (native or Socket.io)	Push order updates to all clients
Routing	React Router v6	Dashboard, order detail, new order pages
Build Tool	Vite	Fast dev server and production builds

7.2 Backend / API (Recommended)

- Node.js + Express or Next.js API routes
- WebSocket server via Socket.io for real-time broadcast
- PostgreSQL or SQLite for order persistence
- REST endpoints: POST /orders, PATCH /orders/:id/status, GET /orders

7.3 Data Models

Order object shape (TypeScript interface):

```
interface Order {
  id: string;
  customerName?: string;
  items: { menuItemId: number; quantity: number }[];
  status: 'new' | 'in_progress' | 'completed' | 'cancelled';
  total: number;
```

```
    createdAt: Date;  
    updatedAt: Date;  
  }
```

8. Screen Inventory

Screen	Route	Key Elements
Dashboard	/	Metrics bar, 4 status columns, real-time order cards
New Order	/orders/new	Menu item selector, quantity picker, customer name, total, submit
Order Detail	/orders/:id	Full item list, status history, advance / cancel actions
Menu Admin	/admin/menu	View 5 items, edit prices (manager only)

9. Out of Scope (v1.0)

- Customer-facing self-ordering kiosk or QR code ordering
- Payment processing or POS integration
- Multi-location / multi-stall support
- Loyalty programme or customer accounts
- Inventory management and stock alerts
- Reporting exports (CSV, PDF) — planned for v1.1
- User authentication / role-based login — planned for v1.1

10. Proposed Milestones

Sprint	Milestone	Deliverables
Sprint 1	Foundation (Week 1–2)	Project scaffold, routing, Tailwind setup, menu data model
Sprint 2	Core Dashboard (Week 3–4)	Order cards, 4-column layout, metrics bar, mock data

Sprint	Milestone	Deliverables
Sprint 3	Real-Time + Backend (Week 5–6)	WebSocket integration, REST API, live order updates
Sprint 4	New Order Flow (Week 7)	Order creation screen, item selector, order submission
Sprint 5	Polish & QA (Week 8)	Responsiveness, accessibility audit, cross-browser testing, UAT

11. Risks & Mitigations

Risk	Likelihood	Mitigation
WebSocket connection drops on poor Wi-Fi	Medium	Automatic reconnect + HTTP polling fallback every 5 s
Order state conflicts (two staff update simultaneously)	Low	Optimistic locking — last write wins with server timestamp
Staff resistance to digital workflow	Medium	Simple UI, < 15-minute training, optional paper backup in week 1
Browser cache serving stale order state	Low	Cache-Control: no-store on API responses + WS invalidation

12. Document Sign-Off

Name	Role	Date	Signature
	Product Owner		
	Tech Lead		
	QA Lead		

— *End of Document* —